

The Arzens

AI Automation & Security Engineering

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Aiman Arshad

Assignment # 03

Task # 01

Security Feature Engineering Report

Part # A

Introduction:

The act of transforming unprocessed security telemetry into useful features that security analysts and machine learning models may utilize to identify cyber threats is known as security feature engineering. Because raw telemetry produced by security technologies frequently contains a lot of data, it might be challenging to directly identify malicious activity. In order to increase the precision and effectiveness of threat detection while protecting user privacy, feature engineering gathers valuable patterns from this data.

Features:

Failed Login Count

Mathematical Definition:

The total number of failed login attempts during a specified time period.

FLC represents the total number of failed login attempts.

Security Detection Purpose:

A high number of failed login attempts may indicate a brute-force attack.

Required Data Sources:

- ❖ Authentication logs
- ❖ Active Directory logs
- ❖ VPN login logs

Privacy Risk Assessment:

Authentication logs may contain usernames, IP addresses, and device identifiers, which are considered sensitive information.

Mitigation Strategy:

- ❖ Hash or pseudonymize usernames.
- ❖ Mask IP addresses where possible.
- ❖ Restrict access to authentication logs.

Login Frequency

Mathematical Definition:

The number of login attempts made within a specified time period.

$LF = \text{No of Login Attempts} / \text{Time Period}$

LF = Login Frequency

Security Detection Purpose:

Attempts to log in too frequently could be a sign of hacked user accounts or automated attacks.

Required Data Sources:

- ❖ Authentication logs
- ❖ Identity management systems

Privacy Risk Assessment:

A user's login history can provide insight into their work patterns.

Mitigation Strategy:

- ❖ Store only necessary login information.
- ❖ Remove unnecessary personal identifiers.
- ❖ Encrypt stored log data.

Network Connection Count

Mathematical Definition:

The total number of network connections established during a specified time period.

$NCC = \text{Total No of Network Connections}$

During a selected observation period.

Security Detection Purpose:

An unusually large number of network connections may indicate malware activity, scanning, or distributed denial-of-service (DDoS) attacks.

Required Data Sources:

- ❖ Firewall logs
- ❖ Network flow records
- ❖ Router logs

Privacy Risk Assessment:

Network logs may expose IP addresses and communication patterns.

Mitigation Strategy:

- ❖ Mask internal IP addresses.
- ❖ Encrypt network logs.
- ❖ Apply strict access control.

Average Packet Size

Mathematical Definition:

The average size of network packets transmitted during a specified time period.

$APS = (\text{Total Packet Size}) / (\text{Number of Packets})$

$APS = \text{Abnormal Packet Size}$

Security Detection Purpose:

Unusual packet sizes could be a sign of virus transmission, data exfiltration, or unusual network behavior.

Required Data Sources:

- ❖ Packet capture (PCAP)
- ❖ Network monitoring tools
- ❖ Firewall logs

Privacy Risk Assessment:

Packet data may contain sensitive communication details.

Mitigation Strategy:

- ❖ Analyze metadata instead of packet contents whenever possible.
- ❖ Encrypt captured data.
- ❖ Limit data retention.

Part # B

Privacy-Aware Feature Engineering:

The goal of privacy-aware feature engineering is to minimize the gathering and usage of personal data while developing features that facilitate security analysis. Sensitive information should be safeguarded throughout the feature engineering process, and only data required for threat detection should be processed.

Pseudonymization:

Pseudonymization replaces direct identifiers such as usernames or email addresses with artificial identifiers.

Example:

- Username: Aiman
- Stored Value: User_006

This reduces the risk of exposing personal identities while still allowing security analysis.

K-Anonymity:

K-Anonymity protects privacy by ensuring that each individual cannot be distinguished from at least **k** other individuals with similar characteristics.

Instead of storing the exact IP address 192.168.1.105, the system may store it as 192.168.1.xxx, making it more difficult to identify a specific device.

Differential Privacy:

Systematically introducing precisely regulated random noise into datasets or statistical conclusions, differential privacy safeguards sensitive data. This keeps information about specific users private while enabling analysts to spot general security trends.

Feature Selection Strategy:

The technique of selecting the best features for security analysis while eliminating unnecessary or redundant data is known as feature selection. By minimizing needless data gathering, effective feature selection lowers computing costs, increases detection accuracy, and protects user privacy.

Examples of useful security features include:

- Failed login count
- Login frequency
- Network connection count
- Average packet size

Conclusion

Security feature engineering transforms raw security telemetry into meaningful features that improve cyber threat detection. Carefully engineered features enable faster identification of suspicious activities such as brute-force attacks, malware communication, and abnormal network behavior.